

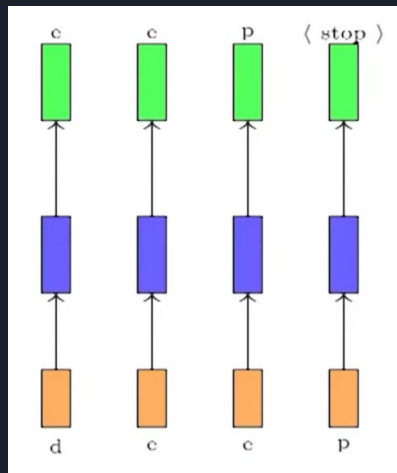
Project 3

Sequence Modelling with RNN and LSTM

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The ML Hub

Sequence Learning Problem

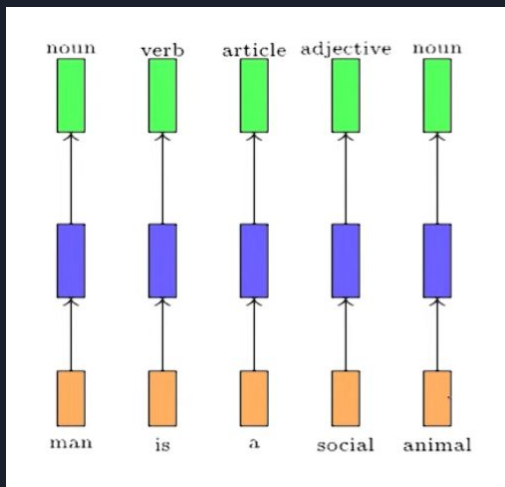
Consider Auto Completion task



- Output of Each Network depends on previous one
- Length of Sequence is not fixed
- Task at each time step is same

Sequence Learning Problem

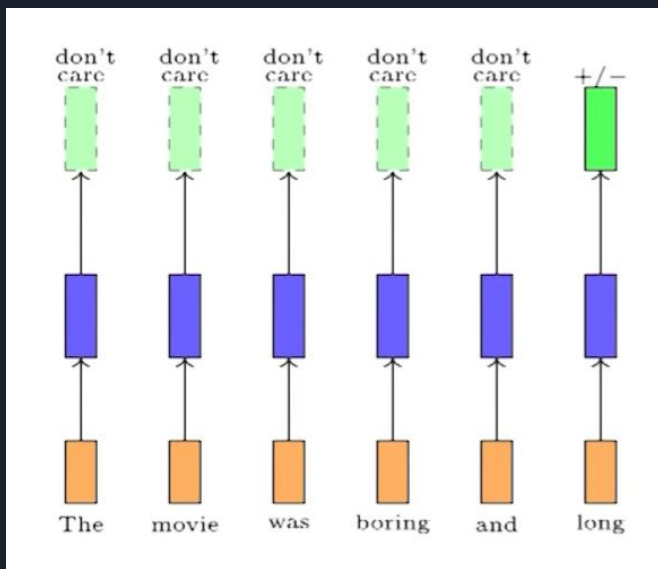
Parts of Speech (PoS) Tagging



- Can be done individually
- Sequential execution performs better
- Noun almost always follows an adjective

Sequence Learning Problem

Sentiment Analysis



- Classify a review as positive or negative
- We can assume that network is doing the same task

the movie was long
but exciting. $\Rightarrow +ve$

Model requirements for Sequence Tasks:

- Takes previous inputs into consideration
- Perform same task at Each time step
- Can take Sequence of Any length

Model requirements for Sequence Tasks:

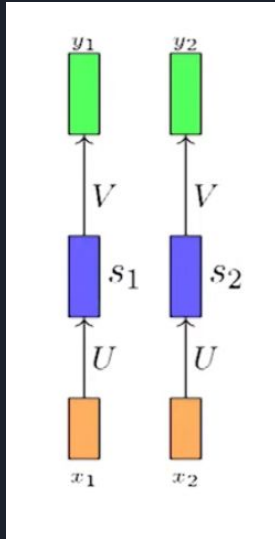
- Takes previous inputs into consideration
- Perform same task at Each time step
- Can take Sequence of Any length



Recurrent Neural Networks

Recurrent Neural Networks

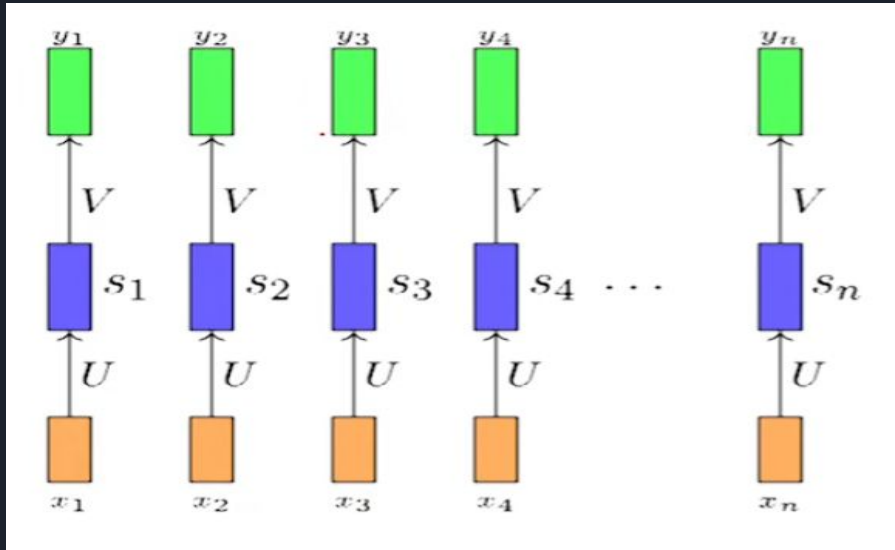
Consider the following structure:



$$x_i \leftarrow \text{input}$$
$$s_i = \sigma(Ux_i + b) \leftarrow \text{hidden represent}^n$$
$$y_i = \phi(Vs_i + c) \leftarrow \text{output}$$

Recurrent Neural Networks

Consider the following structure:



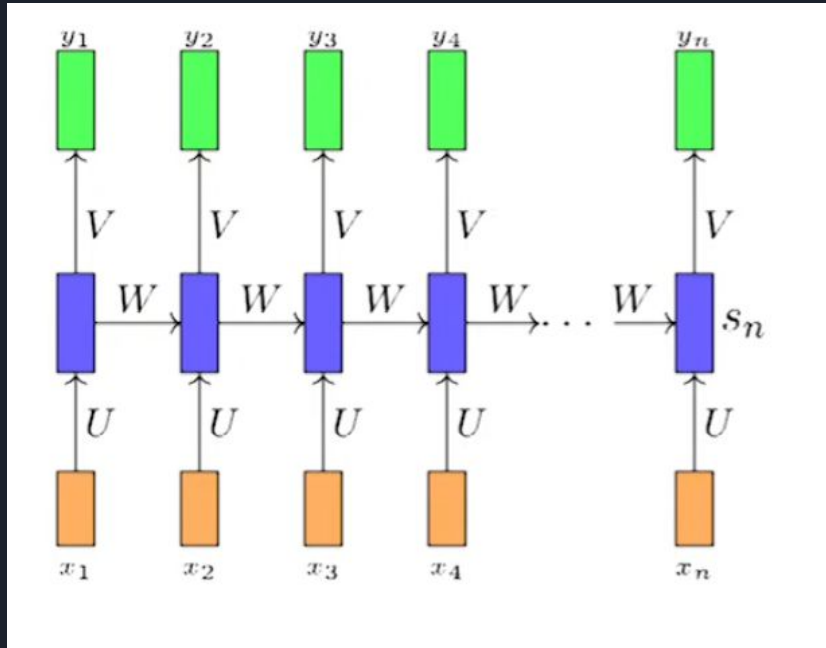
Expand for any number of inputs

Recurrent Neural Networks

$$s_i = \sigma(Ux_i + Ws_{i-1} + b)$$

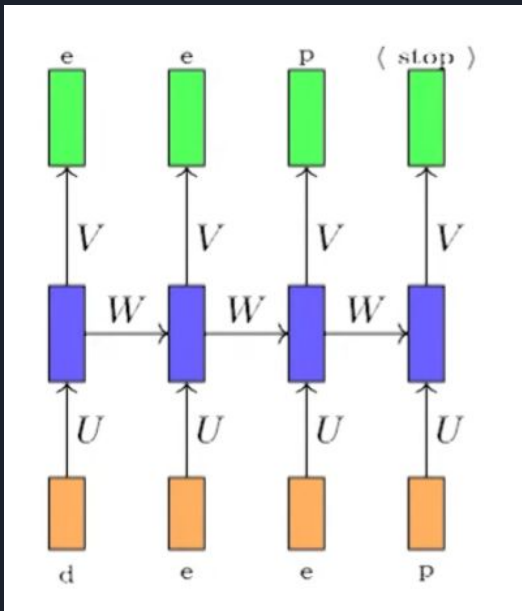
y_i source of previous

Add dependence on Previous Inputs



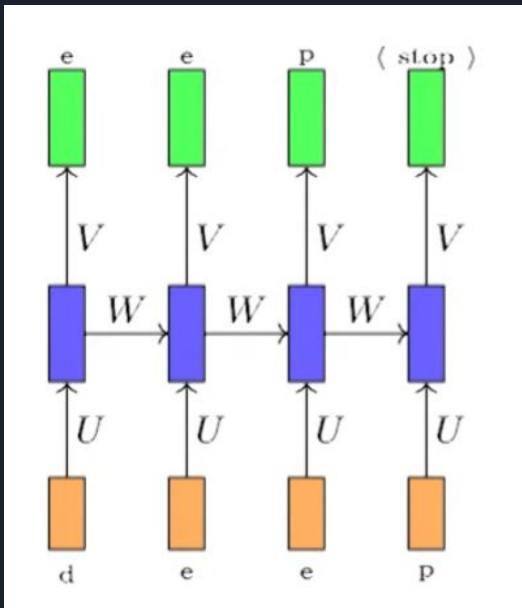
Backprop through Time

Consider next word prediction



Backprop through Time

Consider next word prediction



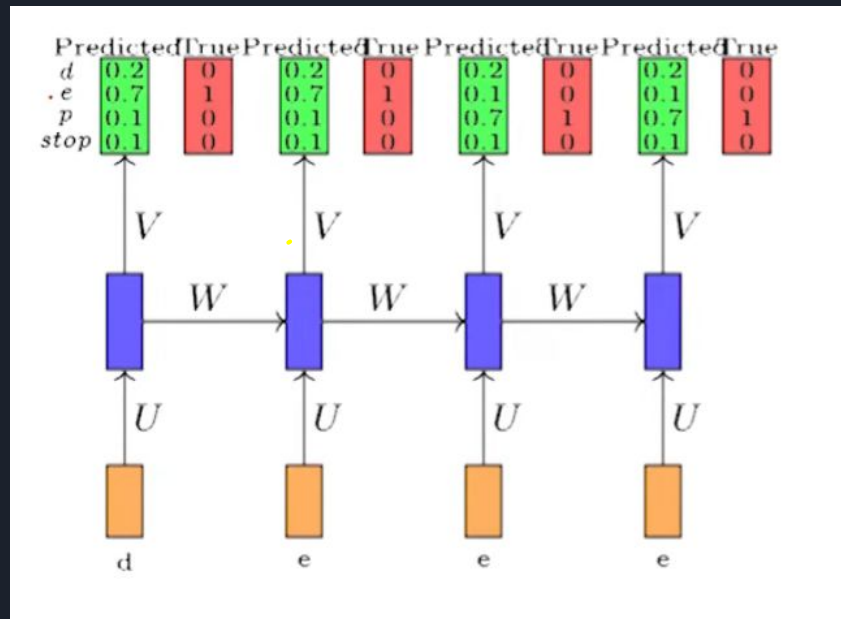
- Softmax Output Function
- Cross-Entropy Loss

Backprop through Time

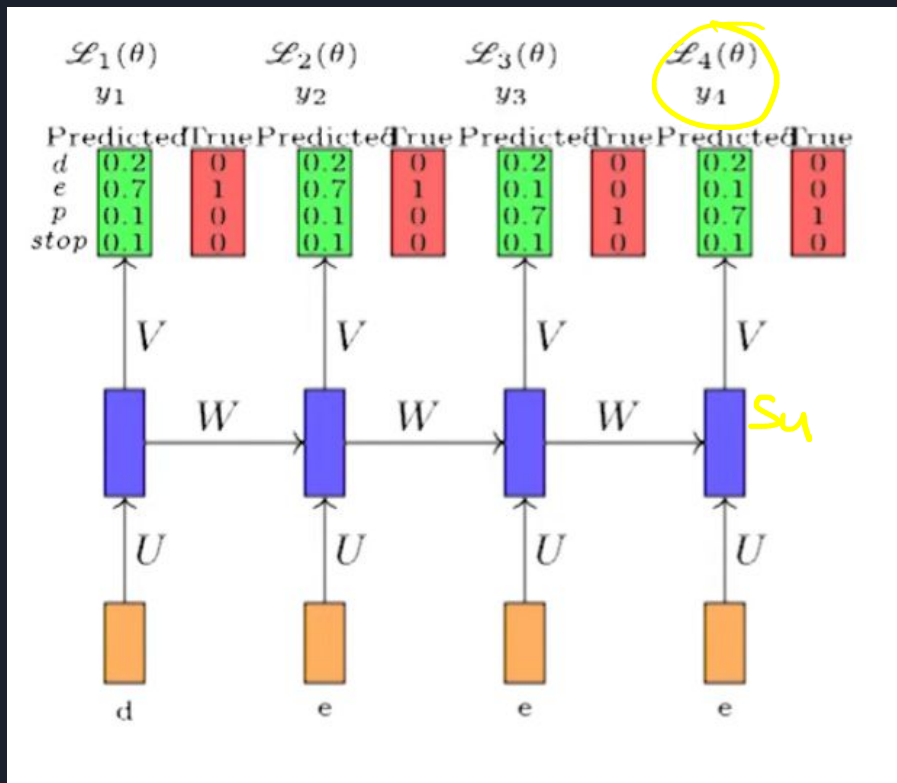
$$s_4 = \sigma(Ws_3 + Ux_4 + b)$$

$s_3 \leftarrow$ also depends on W

Consider next word prediction



Backprop through Time

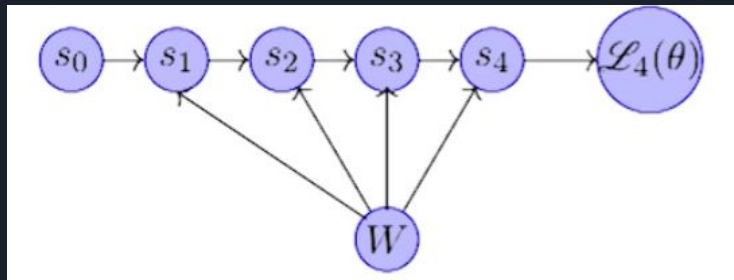


$$L(\theta) = \sum_{i=1}^T \mathcal{L}_i(\theta)$$

$$\frac{\partial L(\theta)}{\partial v} = \sum_{i=1}^T \frac{\partial \mathcal{L}_i(\theta)}{\partial v}$$

$$\frac{\partial L(\theta)}{\partial w} = \frac{\partial L_4}{\partial s_4} \cdot \frac{\partial s_4}{\partial w}$$

Backprop through Time



$$\frac{\partial \mathcal{L}_4}{\partial W} = \sum_{k=1}^4 \frac{\partial \mathcal{L}_4}{\partial s_k} \cdot \frac{\partial s_k}{\partial W}$$

In general

$$\frac{\partial \mathcal{L}_t}{\partial W} = \frac{\partial \mathcal{L}_t}{\partial s_t} \cdot \left(\sum_{k=1}^t \frac{\partial s_t}{\partial s_k} \cdot \frac{\partial s_k}{\partial W} \right)$$

Vanishing and Exploding Gradient

Consider $\frac{\partial S_k}{\partial S_k} = \frac{\partial S_k}{\partial S_{k-1}} \cdot \frac{\partial S_{k-1}}{\partial S_{k-2}} \cdot \dots \cdot \frac{\partial S_{k+1}}{\partial S_k}$

$$\boxed{\frac{\partial S_j}{\partial S_{j-1}}}$$

$$a_j = W S_{j-1} + b, \quad S_j = \sigma(a_j)$$

$$a_j \in \mathbb{R}^d$$

$$a_j = [a_{j1}, a_{j2}, \dots, a_{jd}] \quad \frac{\partial S_j}{\partial S_{j-1}} = \frac{\partial S_j}{\partial a_j} \cdot \underbrace{\frac{\partial a_j}{\partial S_{j-1}}}$$

$$S_j = [\sigma(a_{j1}), \dots, \sigma(a_{jd})]$$

$$\frac{\partial S_j}{\partial a_j} =$$

$$\begin{bmatrix} \sigma'(a_{j1}) & 0 & \dots & 0 \\ 0 & \sigma'(a_{j2}) & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & \dots & \sigma'(a_{jd}) \end{bmatrix}$$

Vanishing and Exploding Gradient

$$\left\| \frac{\partial s_j}{\partial s_{j-1}} \right\| = \left\| \text{diag}(\sigma'(a_j)) w \right\| \leq \left\| \text{diag}(\sigma'(a_j)) \right\| \left\| w \right\|$$

$$\sigma'(a_j) = \sigma(1-\sigma) \leq \frac{1}{4} = \gamma$$

$$\left\| \frac{\partial s_j}{\partial s_{j-1}} \right\| \leq \gamma \left\| w \right\| \leq \gamma \lambda$$

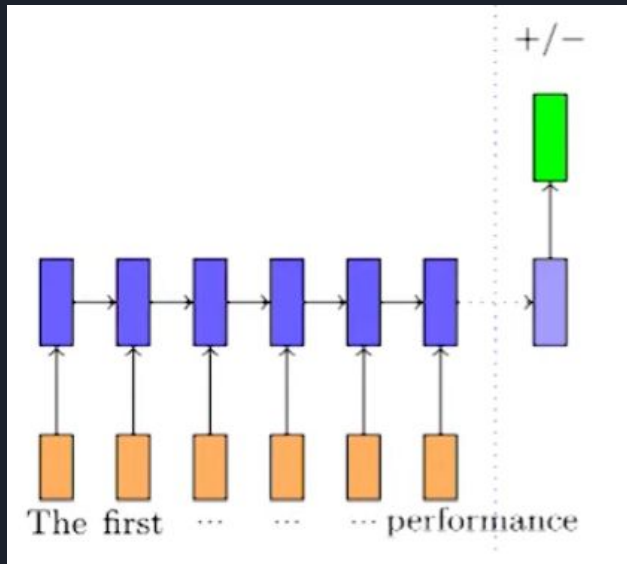
$$\left\| \frac{\partial s_k}{\partial s_1} \right\| = \left\| \prod_{j=2}^k \frac{\partial s_j}{\partial s_{j-1}} \right\| \leq (\gamma \lambda)^{k-1}$$

How do we solve this?

Consider the following movie review:

The first half of the movie was dry but the second half really picked up pace. The lead actor delivered an amazing performance.

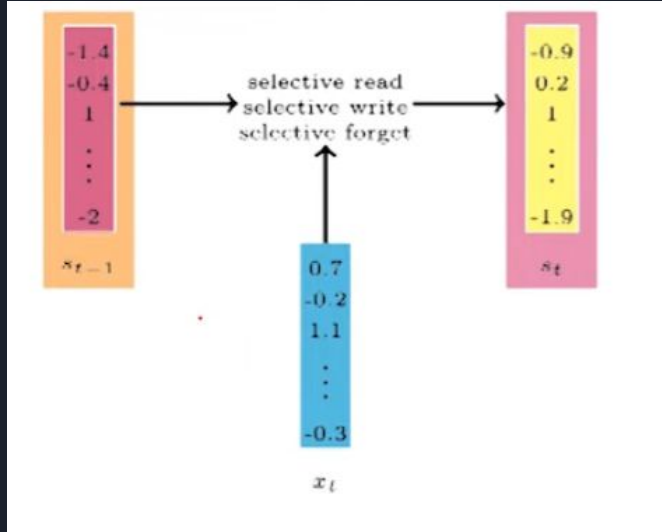
⇒ +ve review



We Want to :

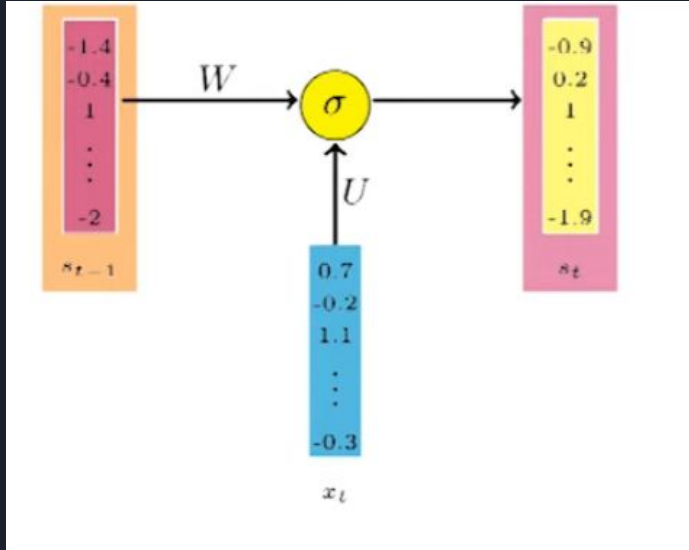
- Selectively Forget (stop words)
- Selectively Read (amazing)
- Selectively Write

Long Short Term Memory (LSTM)



Selective Write

$$s_t = \sigma(Ws_{t-1} + Ux_t + b)$$



$$\begin{bmatrix} 0.1 \\ 0.05 \\ 7.3 \\ 0.7 \end{bmatrix} \xrightarrow[\text{hard}]{\otimes} \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.05 \\ 7.3 \\ 0 \end{bmatrix}$$

$0_t^* \in [0, 1]$

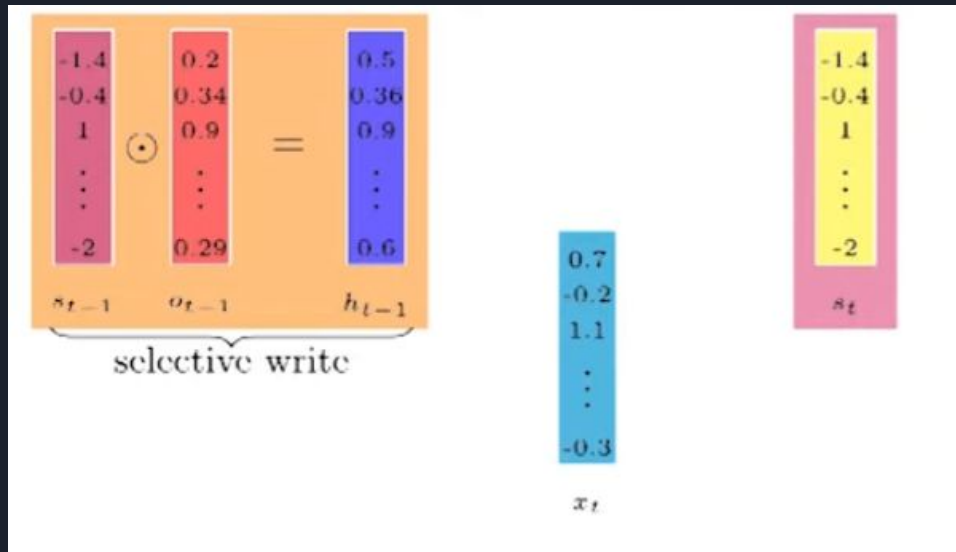
Selective Write

$O_t \leftarrow$ output gate

we compute

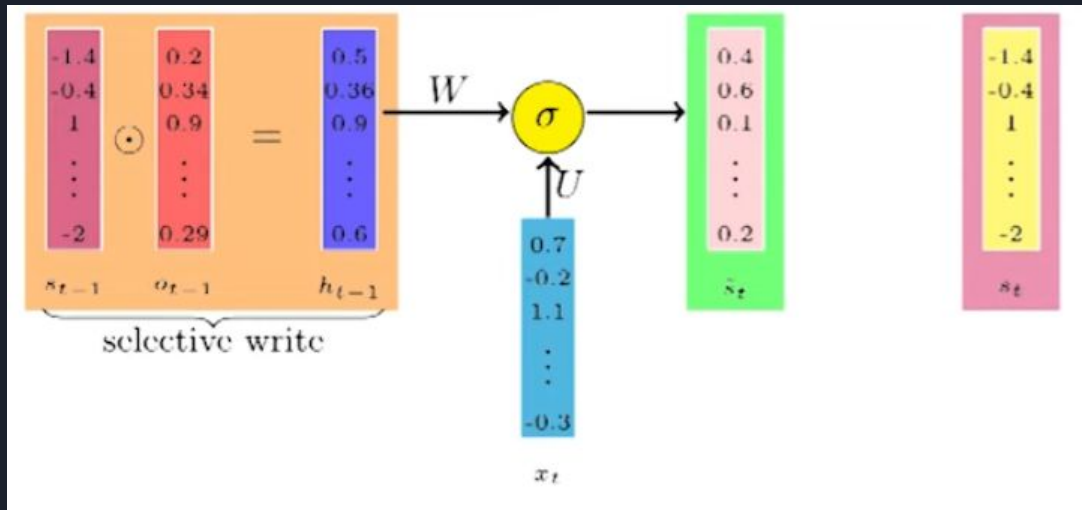
$$h_{t-1} = s_{t-1} \odot o_{t-1}$$

$$o_{t-1} = \sigma(W_o h_{t-2} + U_o x_{t-1} + b_o)$$



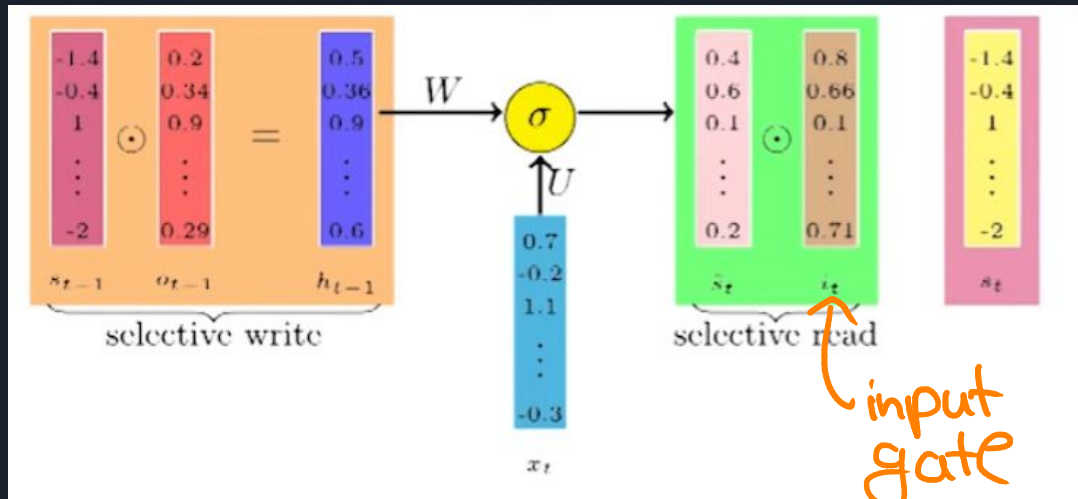
Selective Read

$$\hat{s}_t = \sigma(W h_{t-1} + U x_t)$$



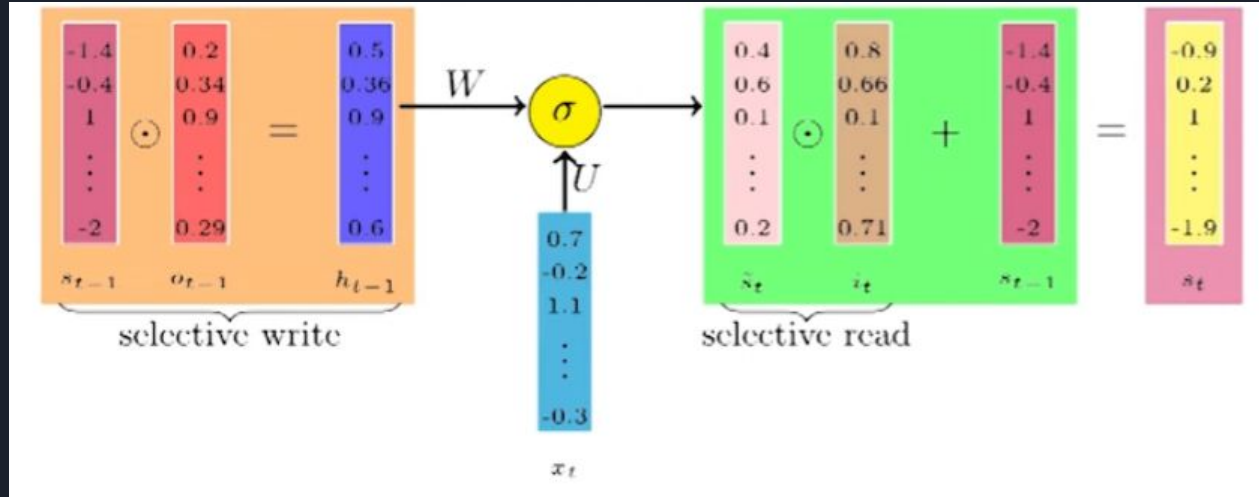
Selective Read

$$i_t = \sigma(W_i h_{t-1} + U_i x_t + b_i) \Rightarrow i_t \odot \tilde{s}_t$$



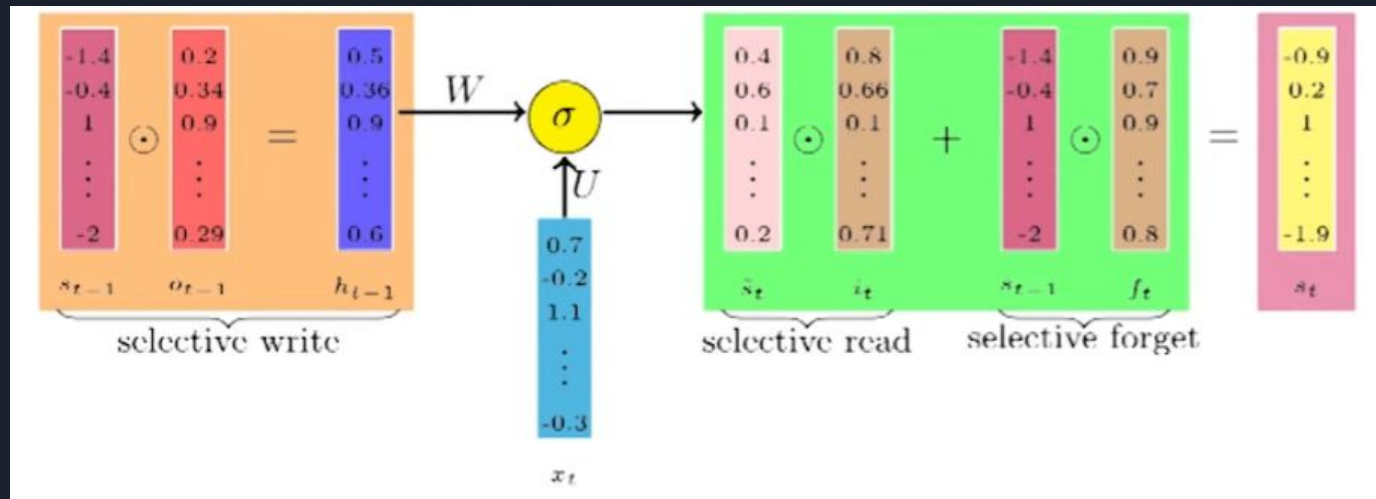
$$s_t \leftarrow (s_{t-1})^{\text{Transform}^t} + f(x_t)$$

Selective Forget



Selective Forget

combine $\tilde{S}_t$ and S_{t-1} to calc. S_t



$$f_t = \sigma(W_f h_{t-1} + U_f x_t + b_f)$$

$$S_t = i_t \odot \tilde{S}_t + f_t \odot S_{t-1}$$



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